

EECS 464: Hands On Robotics - Project 1 Report

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1 Background

1.1 Task Specification

For Project 1 of the Fall 2025 semester of EECS/ROB 464: Hands on Robotics, teams were required to autonomously navigate between a sequence of waypoints within a given space in an allotted time.

Project 0 or P0 of this term was a project that required teams to navigate a track without using wheels or 360 degree turning of any provided motor module. This project resulted in various robots that used legged locomotion to maneuver around the arena. For project 1, teams were encouraged, but not explicitly required, to use the robots from the prior project.

Sensing and waypoint information was provided by the instructional staff by having the teams subscribe to a server that was running on an instructor computer. This server provided information to the robot regarding the sequence of waypoints and the location of the onboard tag. The sequence of waypoints were packaged into a python dictionary and sent via multicast.

1.1.1 Pass/Fail Requirements

To determine if a group would pass in this project, their robot had to reach the final waypoint in the provided sequence while staying in the arena. To determine if the arena boundary was crossed, an AprilTag placed on the robot would be used as a reference for its location, and this tag had to stay within 1 meter of the floor.

1.1.2 Demo Requirements

This project also included two demo milestones that would be used to evaluate the progress of each group. The first demo focused solely on software, as the requirement was that each group could produce a simulated robot that could both respond to commands and update its state information. The second demo applied each group's software to their physical robot, evaluating if the robot could physically respond to direct commands, and then also operate in an autonomous mode.

1.2 Hardware Design

Going into this project, each group was allocated a robot from the selection of robots used in the previous project of this term, and so the hardware design process focused less on making new designs and more on modifying existing ones.

Our team was allocated the project 0 Green team robot, which used 2 gorilla-arm mechanisms to push against the ground and propel itself forward. This motion was performed in a single direction, as a hinge joint in both legs caused the legs to bend and lose their leverage against the ground. This stage was considered the 'resetting' stage of locomotion.

Depending on whether either both arms actuated or just one did, the robot could make a straight or turning motion. Notably, this robot lacked a mechanism for a consistent way of moving backwards.

There were no hardware changes necessary to complete the laid out task, as the project 0 version of the robot was fully capable of multi-directional locomotion, which was all that would be needed to complete the specified task. It should be noted that the location of an AprilTag used for sensing was moved to a different location compared to where it had been for the previous project, though the effect of this change was minimal.



Figure 1: Final robot configuration of the Green robot used for project 1.

1.3 Software Changes

The Project 0 repository for the Green robot required users to call functions in terminal in order to manually control the robot. The robot had three options for motion. Each was defined in its own function. These functions included:

- `move_forward()` - a command that moved both motors forwards together in order to push the robot forwards.
- `turn_right()` - a command that would actuate the left arm only to push the robot towards the right.
- `turn_left()` - a command that would actuate the right arm only to push the robot towards the left.

1.3.1 Motion Characterization

The first thing that was necessary for creating both the robot simulator and running autonomously on the actual hardware was determining how the motion of the system was characterized.

The motion logs for all previous P0 robots were provided in the form of filename.log JSON files. Each entry was timestamped and then provided the relevant tag locations in pixels for each of the tags present in the arena.

```
[0.752454, {"i":16, "p": [[136,351], [133,388], [170,389], [173,353]]}, ...
{"i":26, "p": [[1194,687], [1195,630], [1139,629], [1137,687]]}, ...
{"i":27, "p": [[152,152], [155,103], [106,103], [104,152]]}, ...
{"i":28, "p": [[117,575], [67,575], [67,629], [117,629]]}, ...
{"i":29, "p": [[1201,115], [1147,117], [1148,171], [1204,171]]}]
```

A script was created to parse through each of the time stamped entries from the log file. The script worked by, first, recording the center location of each tag based on its given pixel location. The script would then

filter down the data to create a list of where the center of the robot's tag was in each timestamp throughout the log.

Next, the script would parse through these locations and begin to calculate the difference between the relative x and y locations of consecutive points. This calculated the average velocity between two points. If a given velocity was too low, it was removed from the consideration as it was likely due to the robot not moving, or the resetting action of the legs.

Some final calculations were applied to the list of valid steps to produce a set of statistics that were helpful in categorizing the motion of the robot. These reported metrics included: average stride time, average stride length, the cadence, average velocity, total distance of travel, the total heading drift, and the average heading drift per step.

Table 1: Summary of Gait Parameters

Metric	Value
Average stride time	4.214 s
Average stride length	142.30 px
Cadence	14.2 steps/min
Average velocity	25.54 px/s
Total distance	996.08 px
Total heading drift	6.94 °
Average heading drift per step	1.16 °

When the robot resets the leg configuration to perform another step, it would wobble. This was caused by various points on the bottom of the legs touching the ground. This was recorded in the log files. Filtering this noise proved difficult as if the angle between motions was considered alone, then the wobble would create small circular patterns as it was resetting. These angular changes meant we could not rely solely on the angle between time steps.

The qualification run of the Project 0 Green team provided valuable information regarding the straight line motion of the robot. In this specific example, all the motion by the robot was a straight line step. This allowed for calibration of the script. Although the script was not perfect in its ability to identify individual steps and turns, it helped provide an estimate of the motion of the robot.

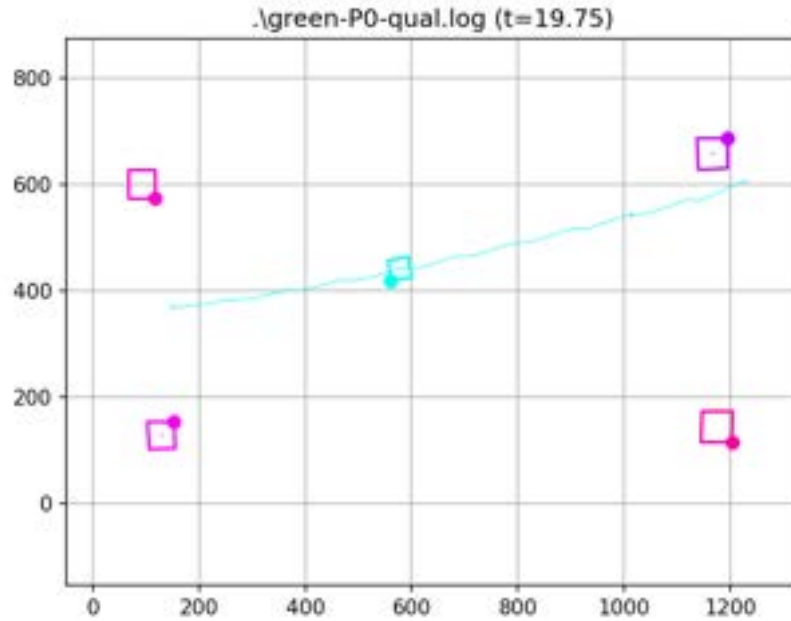


Figure 2: The qualification trial of the project 0 Green team reveals that, even though, the motion of the robot is straight, it does contain some heading drift in the system.

A massive benefit of computing these motion metrics would be that it would help create an autonomous simulation that would better represent the actual motion of the physical robot.

1.3.2 Expanding the sensor range

The first attempt at expanding the sensor range was by streaming a new point of view of the AprilTag using an off-board phone. By seeing the AprilTags from a new point of view, the relative location of the robot to the tags could be calculated. This was done by using the AprilTag iOS app developed at the University of Michigan, which had a feature built-in to enable streaming the tag ID's and corner locations over UDP. The Green team decided not to pursue this route due to the added complexity of fitting the tags and converting the coordinate frames.

The second attempt to gain new information was by exposing the sensor suite on an Android device. The idea was that the phone could be attached to the robot and could aid in informing the heading angle of the robot. By using the accelerometer on the phone to track heading, the target direction and relative direction of the robot could be calculated. This method used an open source app to expose the sensor reading from an Android device to a listening WebSocket. This was also not pursued due to the added complexity of having to interpret the data and then apply it to the robot actions.

1.3.3 Final Software Design

The final controller on the physical robot operated using a bang-bang feedback scheme that converted the nearly-real-time sensor input from the instructor's vision system into discrete movement commands. Two virtual sensors, located on the right and left sides of the robot's AprilTag (rotated 90 degrees CW from convention), measured perpendicular distance to the line between the current and next waypoints. When both readings were high, the robot advanced forward using synchronized actuation of both arms. When one

sensor dropped, the controller issued a single-arm turn toward the weaker side to correct heading. Sensor readings ranged from 0 to 255, where 0 indicated that the sensor was far from the waypoint line segment (or off of it) and 255 indicated that it lay directly on the line. The system contained configurable thresholds for interpreting these values: a high threshold of 50 on this scale defined when a sensor was treated as “on the path,” while differences greater than 30 between sensors triggered a heading correction. A lower bound of 20 defined the “dead zone,” where both sensors were considered too weak to produce reliable direction information and the robot maintained its previous heading.

2 Results

2.1 Overall Team Performance

Throughout all three trials, the Green Team’s robot showed steady progress in both control and accuracy through shorter times taken to reach waypoints.

In trial one, the robot’s path was uneven, with wide turns and loops as it tried to find the correct direction. In this first trial, the robot managed to reach to the first waypoint, but when crossing over it, the robot did not register the april tag crossing the waypoint’s position and continued to move in a straight-line path towards where it calculated the waypoint was.

By trial two, movement appeared more stable and direct, with fewer sudden heading shifts. The robot successfully reached waypoint one and began moving toward waypoint two. On the way there, however, it deviated from the path towards the waypoint as it approached and eventually exited the arena and disqualified the attempt.

In the third trial, the robot attempted turning adjustments to approach the waypoint but again lost its location data and deviated from the intended path. The robot was unable to make corrections needed with its large strides and turn angles.

Across all three runs, the robot demonstrated incremental improvement, following straighter paths and maintaining better directional stability. Although it did not complete all waypoints, performance reflected stronger synchronization between sensor feedback and motion execution.

2.1.1 Blue Team Results

The Blue Team’s robot completed the first trials autonomously and reached all waypoints. It stayed within the arena and followed the waypoint path smoothly. In the first trial, the robot maintained a stable heading with consistent motion. In the second trial, the robot demonstrated a spiral-like pattern after reaching the first waypoint and couldn’t recover. In the third trial, it showed even smoother movement and more direct transitions between waypoints, however failed to reach the final waypoint. This was the only robot to fully complete the challenge autonomously within the 15 minute requirement. Overall, the Blue Team’s robot demonstrated precise and reliable autonomous performance.

2.1.2 Red Team Results

The Red Team’s robot completed its trial manually after partial autonomous movement. It reached the first waypoint but drifted significantly when navigating toward the next, stopping near the arena’s center. The robot remained within bounds but did not complete the full sequence.

2.1.3 Maize Team Results

The Maize Team’s robot reached the first waypoint in all trials but did not complete the sequence. It stayed within the arena and showed consistent starts but lost stability midway through each run. Overall, the robot demonstrated partial success with limited waypoint completion.

3 Discussion

Although all teams successfully met the pass/fail requirements, there were multiple teams that had to switch to manual mode to complete the task. Each of the teams’ designs had their own merits and shortcomings, which contributed to the results shown above. When it came to following the line set between the current two tags in the waypoint list, which we will refer to as the desired path, each robot would end up straying away from this desired path a varying amount, and the farther each robot got from the desired path, the faster that team would have to resort to switching to manual mode.

3.1 Blue Team

The blue team’s robot was the only robot that was capable of completing the full waypoint list while under autonomous control. The robot consisted of a box design with 2 wheels placed on each side of the box. Notably this team was the only team to design a fresh robot instead of modifying an old one. The wheel-based locomotion set this robot apart from all other teams’ robots as the blue robot was able to make very small steps, a feature that worked to their advantage as in their successful run the robot never strayed too far from the desired path. The downside to this approach was that this robot was extremely slow to complete the trial, almost being disqualified for exceeding the time limit, though with further testing they may be able to find a step size that keeps the stability of the smaller step sizes while reducing the time taken to complete the task.

3.2 Red Team

The red team’s robot was a slightly modified version of the project 0 maize team’s robot, keeping intact the leg-kicking based locomotion while stripping auxiliary parts such as the tail and modifying inconsistent parts such as the wheel. The maize robot was quick to fail when it came to its autonomous navigation, and there appeared to be a few factors working against the success of this robot. First was the new wheel design, which ended up having a problem with slipping on the ground, reducing the change in heading of any turn command. This mismatch in expected heading change compared to observed heading change resulting in the robot straying from the desired path to reach the first waypoint, and upon getting too far away the team resolved to switch to manual control. The second factor working against this robot was the fixed large step size. Due to the locomotion profile of this robot relying on a kick motion, the robot was forced to execute long step lengths in the arena, which meant that any errors in heading would be amplified as the forward motions would set the bot much further away from the desired path in a single command. Unfortunately for this group, these issues are tied to the key features of the design of the maize robot, and so we believe that a way to improve the effectiveness of this robot would be to have major design changes. Such changes could include changing wheel materials to improve traction or shortening the leg lengths to reduce step sizes.

3.3 Maize Team

The maize team inherited the project 0 blue team’s robot, though they decided to overhaul the motion to rely less on the rotational inertia of a spinning bottom plate on the robot and work more with two small gorilla-like arms in the front. This robot had a solid turning ability as it was able to take advantage of a 360 degree motor attached to the bottom plate, making the robot able to reach exact headings instead of being forced to use fixed increments when changing direction. Similar to the red team, there were some inconsistencies in the friction between the carpet and the cardboard that did result in some slippage, though there was a lesser effect on this robot from the slippage. Due to some inconsistencies in the two gorilla arms,

there was a slight drift in the robot's trajectory when moving forward, and this drift caused the robot to fall off of the desired path and mess up the autonomous navigation.

3.4 Green Team

Our team inherited the project 0 Green team's robot, and used it to complete the project 1 task with minimal modifications. While the Green robot had a solid level of success in project 0, the same cannot be said about the performance in this project, and it can be said that the strengths of the robot for project 0 ended up working as weaknesses in project 1. The stride length and fixed turn angle of this robot led to problems for this project, as the robot struggled to maintain the correct heading to follow the desired path, and the small discrepancies in the heading would be amplified by the large stride length, leading to the robot missing waypoints. Another critical flaw in the design of this robot was the inconsistency between the two gorilla arms and each pivot motor attached to the arms, which would lead to a drift in the trajectory when moving forward, similar to that of the Maize team's robot. Likewise, this drift would end up throwing the Green robot off path and the previously mentioned fixed turn angle meant that these small discrepancies were hard to adjust for. On the software end, the robot's autonomous control relied on a bang-bang algorithm, which followed the desired path initially but would have trouble recovering once the error caused by the drift caused the sensors to read 0. The flawed recovery behavior of the robot meant that once we had hit this critical point of 0 on sensor readings, the robot would be switched to manual mode to complete the trial. The poor recovery also resulted in a poor transition between waypoints, as for the condition of hitting a waypoint to register one of the two sensor readings would be lost due to the relation between the tag and the waypoint at this transition, and as a result our robot failed to clear more than one waypoint autonomously.

3.4.1 Future Improvements

If we were to make future iterations on this robot, we would heavily consider both replacing all of our motors and rebuilding our legs to increase the consistency in the forward motion of this robot. To go along with these changes, we would also test our robot on the material used in the arena so that we could know if there still existed any drift so that we may adjust our software to further account for any error caused by the drift. Testing the recovery behavior would be highly beneficial for future software improvements, as this was a major weakness shown through all of the P-day trials.